

Maximum Load in n-Balls and n-Bins experiment

As asked, we perform the *n-Balls and n-Bins* experiment for $K = 1000$ times and compute the maximum load of a bin inside the bins set.

After that, we plot the noted maximum loads in the same figure as the Lower and Upper Bound, given by $L = \frac{\ln n}{\ln \ln n}$ and $M = 3 \frac{\ln n}{\ln \ln n}$, respectively.

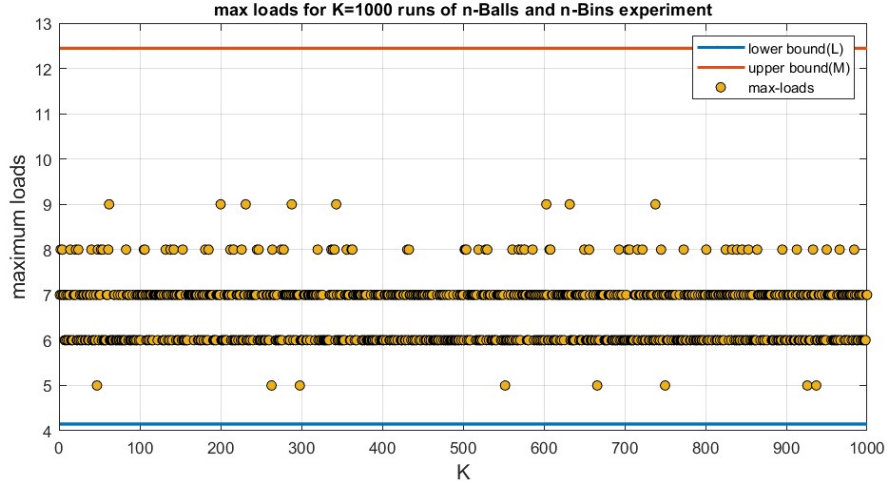


Figure 1: max loads for the $K = 1000$ iterations of the n-Balls and n-Bins experiment

When it comes to the upper bound M of the maximum loads, we can clearly see that for the chosen n , which is sufficiently large, the Lemma regarding the upper bound of the experiment states that, M will be an upper bound of the maximum load, with probability:

$$\Pr(\text{max_load} > 3 \frac{\ln n}{\ln \ln n}) \leq \frac{1}{n} = \frac{1}{10^4} = 0.0001$$

which is very low, implying that M will be a very loose upper bound.

The above statement can be confirmed just by checking the plot of maximum loads, where it is obvious that no measurement out of the $n = 10^4$ made, exceeds or even reaches M

(thick red line).

On the other hand, when it comes to the lower bound of the maximum loads, the respective Lemma guarantees, that for sufficiently large n :

$$\Pr(\text{max_load} > \frac{\ln n}{\ln \ln n}) \geq 1 - \frac{1}{n}$$

which in our case of study ($n = 10^4$) gives $\Pr(\text{max_load} > \frac{\ln n}{\ln \ln n}) \geq 1 - \frac{1}{10^4} \approx 1$.

This implies that with great probability, L is a very tight lower bound to the *n-Balls and n-Bins* experiment. This is quite obvious from looking at the plot, where the lowest of maximum loads greatly approach L (thick blue line).

Hamiltonian Cycles

For the described experiment to be executed, p was chosen to be exactly $\frac{40 \ln n}{n}$, giving a value for q from the formula $q = 2q - q^2 \Leftrightarrow q = 1 - \sqrt{1 - p}$.

In order to test the correctness of our implementation, at the final step of the experiment, we use the *verifyHC*(path) function to check whether or not the passed path, which the modified algorithm returned as a certified Hamiltonian Cycle, is indeed an HC.

Sure enough, after following the detailed steps of the description and manually running the experiment many times, we can conclude that the modified HC Algorithm is very unlikely to fail, given a suitably prepared graph G of n nodes ($G_{n,p}$).

If the modified algorithm was successful, *verifyHC*(path) prints the following in the MATLAB command window:

There are n distinct vertices in the path.

The first and last vertices of the path match!

The given HP, is indeed an HC!

thus confirming the correctness of the algorithm.

Lastly, it is important to note that our results very much align with what the Corollary in our theory states, which is that if for suitably selected parameters p, q a $G_{n,p}$ graph is created and given as an input to the modified HC algorithm, an HC will be found with very high probability ($1 - O(\frac{1}{n})$).